CISC 473: Deep Learning — Project Proposal

P08: Abstractive Summarization via Parameter-Efficient Fine-Tuning

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GitHub Repository: https://github.com/wenbin05/DeepLearningP08

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Problem Statement and Motivation

Abstractive text summarization aims to generate concise and coherent summaries that capture the important information in a source document using newly generated language, rather than simply extracting sentences from the original text. This task is challenging because a good summary must simultaneously identify salient information, preserve the meaning of the source, remain linguistically fluent, and avoid introducing unsupported or contradictory facts. Modern encoder–decoder Transformer models such as BART have achieved strong performance on abstractive summarization by leveraging large-scale pre-training before being adapted to downstream datasets [1].

However, adapting large pre-trained models through conventional full fine-tuning requires updating most or all of their parameters. As model sizes increase, this creates substantial computational, memory, and storage costs, especially when separate versions of the same model must be maintained for different tasks or domains. Parameter-efficient fine-tuning (PEFT) methods address this problem by keeping most of the pre-trained model fixed and training only a small number of additional or modified parameters. Methods such as Low-Rank Adaptation (LoRA) [2] and Prefix Tuning [3] have therefore become important approaches for reducing the cost of adapting pre-trained Transformer models.

Despite their efficiency, an important question remains: how should limited adaptation capacity be used while preserving high-quality summarization performance? Reducing the number of trainable parameters may improve efficiency, but it can also affect summary quality, factual consistency, and the ability to generalize to text from a different domain. These trade-offs are important in practical applications such as news summarization, document analysis, and information retrieval, where models may need to be adapted repeatedly under limited computational resources. This project therefore studies parameter-efficient adaptation of BART for abstractive summarization, with particular attention to the balance between efficiency, summarization quality, factual reliability, and cross-domain generalization.

Literature Survey

Related Works

BART: Lewis et al., 2020 [1]. Lewis et al. introduced BART, a denoising sequence-to-sequence pre-training framework that combines a bidirectional Transformer encoder with an autoregressive decoder. The model is pre-trained by corrupting text and learning to reconstruct the original sequence, allowing it to perform strongly on conditional generation tasks such as abstractive summarization. On the CNN/DailyMail benchmark, BART-large achieved ROUGE-1/2/L scores of

44.16/21.28/40.90, while on XSum it achieved 45.14/22.27/37.25 [1]. These results established BART as a strong summarization architecture and motivate its use as the backbone of our project. However, Lewis et al. adapted BART through conventional full fine-tuning, whereas our project investigates whether comparable summarization performance can be obtained by training only a small fraction of the model parameters.

LoRA: Hu et al., 2022 [2]. Hu et al. proposed Low-Rank Adaptation (LoRA), which freezes a pre-trained weight matrix W_0 and represents its task-specific update using two much smaller trainable matrices such that $\Delta W = BA$. This reduces the number of parameters that must be optimized while allowing the learned update to be merged into the original model at inference time. On GPT-3 175B, a LoRA configuration with 4.7 million trainable parameters achieved ROUGE-1/2/L scores of 53.8/29.8/45.9 on the SAMSum summarization dataset, compared with 52.0/28.0/44.5 for full fine-tuning of approximately 175 billion parameters [2]. LoRA is the main PEFT method studied in our project. In contrast to the original paper's broad evaluation across several models and tasks, we focus specifically on BART-based abstractive summarization and study how LoRA rank and placement influence performance, efficiency, and transfer.

Prefix-Tuning: Li and Liang, 2021 [3]. Li and Liang proposed Prefix-Tuning, a parameter-efficient alternative to full fine-tuning in which the pre-trained model remains frozen while a small set of continuous task-specific prefix vectors is optimized. These learned vectors act as virtual tokens that subsequent Transformer activations can attend to. Using BART-large on XSum, full fine-tuning achieved ROUGE-1/2/L scores of 45.14/22.27/37.25, while Prefix-Tuning achieved 43.80/20.93/36.05 using a 2% configuration and 42.92/20.03/35.05 using approximately 0.1% task-specific parameters [3]. The authors also found stronger extrapolation to unseen topics in some settings. Prefix-Tuning therefore provides an important PEFT baseline in our project because it adapts the model through learned conditioning vectors, rather than through low-rank modifications of its weight matrices as in LoRA.

AdaLoRA: Zhang et al., 2023 [4]. Zhang et al. observed that assigning the same low-rank adaptation budget to every weight matrix implicitly assumes that all model components are equally important, and proposed AdaLoRA to allocate adaptation capacity dynamically according to parameter importance. AdaLoRA uses an SVD-inspired parameterization and progressively removes less important rank components while preserving more capacity in influential matrices. With BART-large and a 1.10% trainable-parameter budget, AdaLoRA achieved ROUGE-1/2/L scores of 44.35/21.13/36.13 on XSum, compared with 43.40/20.20/35.20 for standard LoRA. On CNN/DailyMail, the corresponding scores were 44.96/21.77/42.09 and 44.72/21.58/41.84 [4]. This work is closely related to our project because it demonstrates that the allocation of a fixed adaptation budget can affect performance. Our study differs by examining explicit and interpretable architectural locations within BART rather than learning the allocation dynamically.

WeightLoRA: Veprikov et al., 2026 [5]. Veprikov et al. proposed WeightLoRA, which treats individual LoRA adapters as selectable units and learns which adapters are sufficiently important to retain during fine-tuning. Their WeightLoRA+ variant reallocates the saved parameter budget by increasing the rank of selected adapters. In BART-large experiments with rank $r = 8$, standard LoRA used approximately 1.05% of the model parameters and obtained ROUGE-1 scores of 36.07 on XSum and 25.28 on CNN/DailyMail, whereas WeightLoRA+ used the same reported parameter percentage and obtained 36.74 and 25.72, respectively [5]. These results provide recent

evidence that the usefulness of LoRA adapters is not uniform throughout a model. Our project differs from WeightLoRA because we do not learn which individual adapters to retain; instead, we perform a controlled comparison of architecturally distinct BART regions under approximately matched trainable parameter budgets and additionally examine cross-domain transfer and factual consistency.

FactCC: Kryscinski et al., 2020 [6]. Kryscinski et al. argued that lexical-overlap metrics such as ROUGE do not directly determine whether a generated summary is factually consistent with its source and introduced FactCC, a weakly supervised factual-consistency classifier for abstractive summarization. The method generates synthetic consistent and inconsistent claims from source documents and trains a BERT-based classifier to determine whether a claim is supported by the source. FactCC achieved 72.65% class-balanced accuracy and a weighted F1 score of 0.86 on its annotated CNN/DailyMail evaluation set, but its balanced accuracy decreased to 54.11% on XSum [6]. This result is relevant to our project because it shows both the importance and difficulty of factuality evaluation across domains. We therefore evaluate factual consistency in addition to ROUGE and retain human evaluation rather than relying exclusively on an automatic factuality metric.

Taken together, these studies establish the progression motivating our work. BART provides a strong encoder–decoder architecture for abstractive summarization, while LoRA and Prefix Tuning demonstrate that downstream adaptation can be performed with far fewer trainable parameters than full fine-tuning. AdaLoRA and WeightLoRA further indicate that adaptation capacity is not equally useful across all model parameters or adapters. At the same time, FactCC shows that factual consistency cannot be assessed solely through lexical-overlap metrics such as ROUGE. These findings motivate a controlled study of where a limited LoRA budget should be placed within BART and whether that choice affects not only ROUGE performance but also cross-domain generalization and factual consistency.

Open Problem

AdaLoRA [4] and WeightLoRA [5] indicate that adaptation capacity need not be distributed uniformly across Transformer parameters. However, these works do not provide a controlled comparison, in BART-based abstractive summarization, of encoder self-attention, decoder self-attention, and encoder–decoder cross-attention under approximately matched trainable-parameter budgets.

This distinction is important because these components perform different roles in an encoder–decoder model. A direct comparison without controlling parameter count would be confounded: adapting more modules generally introduces more trainable parameters, making it unclear whether a performance improvement is caused by the architectural location of adaptation or simply by additional capacity.

Our project addresses this gap by comparing LoRA adaptation across these architectural regions while keeping their trainable-parameter counts approximately matched. We will investigate whether the placement of a limited LoRA budget affects in-domain summarization quality, zero-shot transfer from CNN/DailyMail to XSum, factual consistency, and training efficiency.

Proposed Methodology

Baseline

We will use the pre-trained `facebook/bart-base` checkpoint from HuggingFace Transformers as our backbone model and the standard CNN/DailyMail train, validation, and test splits as the primary summarization benchmark. A zero-shot BART-base model will first be evaluated to establish pre-adaptation performance. We will then implement the three required adaptation baselines: full fine-tuning, Prefix Tuning, and Low-Rank Adaptation (LoRA), using HuggingFace Transformers and the PEFT library.

For LoRA, we will evaluate ranks $r \in \{4, 8, 16\}$ and perform the required target-projection ablation over Q, K, V, and all selected attention projections. Summarization quality will be measured using ROUGE-1, ROUGE-2, and ROUGE-L. Training configurations will use the same preprocessing, generation settings, and data splits wherever possible so that differences between adaptation methods can be compared fairly.

Proposed Extension

Our extension investigates whether the *architectural location* of LoRA adaptation affects summarization performance when trainable capacity is held approximately constant. We will construct three placement variants in which LoRA is applied only to (1) encoder self-attention, (2) decoder self-attention, or (3) encoder–decoder cross-attention. The same target projections, LoRA rank, and training hyperparameters will be used across these variants. Because BART-base contains the same number of layers and hidden dimensionality in these corresponding attention blocks, this design yields approximately matched trainable-parameter counts and isolates placement as the main experimental variable.

We chose this extension because encoder self-attention, decoder self-attention, and cross-attention perform different roles: representing the source, modeling generated text, and conditioning generation on the source, respectively. Our hypothesis is that, even under similar parameter budgets, LoRA placement will lead to measurable differences in summarization quality and cross-domain robustness. This would indicate that parameter efficiency depends not only on how much adaptation capacity is used, but also on where that capacity is introduced.

Evaluation Plan

Models will be trained on CNN/DailyMail and evaluated on its held-out test set. To assess cross-domain generalization, the same trained models will also be evaluated on the standard XSum test split without further fine-tuning.

We will report ROUGE-1, ROUGE-2, and ROUGE-L for summarization quality, together with the number and percentage of trainable parameters and training time for efficiency. Factual consistency will additionally be evaluated using the required human study of 10 participants over 20 generated summaries. Comparisons will include zero-shot BART-base, full fine-tuning, Prefix Tuning, standard LoRA, and our three LoRA placement variants. Main configurations will be repeated with at least three random seeds, and results will be reported as mean $\pm$ standard deviation.

Early Novel Contribution (Optional Bonus)

Our proposed novel contribution is a budget-matched ablation of LoRA placement across BART's encoder self-attention, decoder self-attention, and encoder–decoder cross-attention. By fixing the target projections, LoRA rank, and training settings, we isolate architectural placement as the main

variable rather than confounding it with trainable capacity. We will test whether placement affects in-domain ROUGE performance, zero-shot CNN/DailyMail-to-XSum transfer, factual consistency, and training efficiency.

Project Timeline and Roles

Week	Tasks	Responsible
1	Read BART, LoRA, Prefix-Tuning, AdaLoRA, WeightLoRA, and FactCC; set up GitHub repository and Python environment; load CNN/DailyMail and XSum.	Both
2	Finalize research question, placement experiment, and evaluation protocol; implement dataset preprocessing and ROUGE evaluation pipeline; complete proposal. **(Proposal due)**	Both
3	Evaluate zero-shot BART-base on CNN/DailyMail; verify generation and ROUGE pipeline; implement parameter-counting and experiment logging utilities.	Wen Bin / Junn Rong
4	Implement LoRA using HuggingFace PEFT; run rank experiments $r \in \{4, 8, 16\}$ and begin $Q/K/V$/all target-projection ablations.	Wen Bin / Junn Rong
5	Complete LoRA target-projection ablations; implement and evaluate Prefix Tuning and full fine-tuning baselines under comparable training budgets.	Junn Rong / Wen Bin
6	Evaluate trained models on XSum without further fine-tuning; compile preliminary quality, efficiency, and cross-domain results; perform initial analysis and write midterm report. **(Midterm due)**	Both
7	Implement encoder-only, decoder-only, and cross-attention LoRA placement variants; verify approximately matched trainable-parameter counts; prepare and conduct the human factual-consistency evaluation.	Wen Bin / Junn Rong
8	Complete placement experiments with multiple random seeds; analyse human factuality results; measure training time and parameter efficiency; conduct extended ablations where required.	Both
9	Aggregate mean $\pm$ standard deviation results; produce ROUGE-versus-parameter trade-off plots and placement comparison tables; perform failure analysis; draft final report and clean the codebase.	Both
10	Finalize report, figures, README, requirements, and single-command reproduction workflow; verify results from a clean environment and prepare the final presentation. **(Final due)**	Both

Contribution Declaration

Member	Primary Responsibilities	Estimated %
Choo Wen Bin	Project Lead; experiment design; LoRA implementation; architectural-placement experiments; experiment integration; statistical analysis; report coordination and final submission.	50%
Nai Junn Rong	Research Lead; literature review; dataset and evaluation pipeline; Prefix Tuning and full fine-tuning baselines; human factual-consistency study; reproducibility checks and documentation.	50%

References

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[5] Veprikov, A., Solodkin, V., Alexander, Z., Savchenko, A., and Beznosikov, A., 2026. "Weight-LoRA: Keep Only Necessary Adapters." *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers*, pp. 12419–12437. DOI: 10.18653/v1/2026.acl-long.566.

[6] Kryscinski, W., McCann, B., Xiong, C., and Socher, R., 2020. "Evaluating the Factual Consistency of Abstractive Text Summarization." *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9332–9346. DOI: 10.18653/v1/2020.emnlp-main.750.